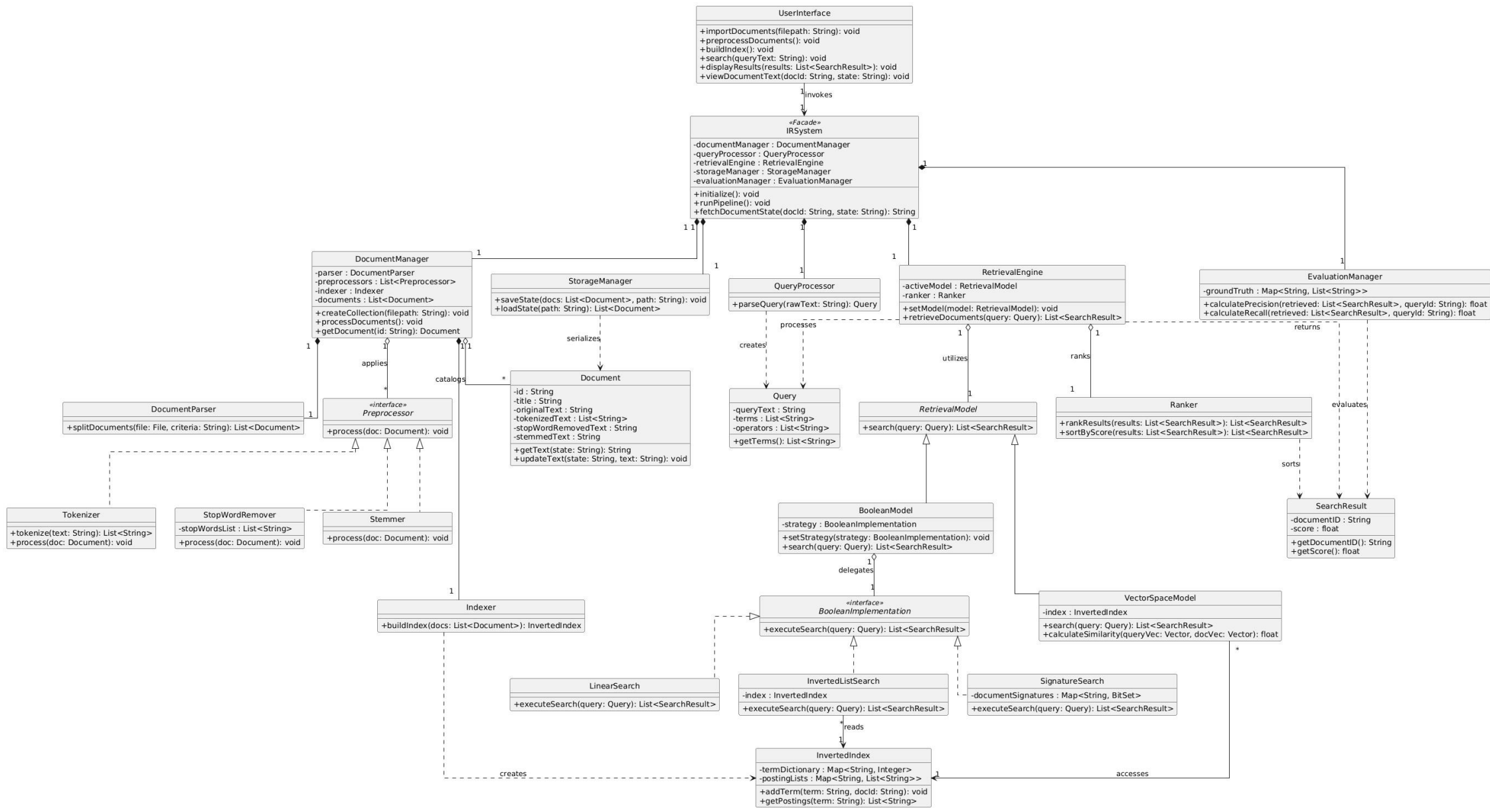


Task Sheet 1

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Explanation of System Components

1. UserInterface

The UserInterface class represents the interaction layer between the user and the Information Retrieval system. It allows users to import documents, trigger preprocessing and indexing, submit search queries, display search results, and view document contents after different preprocessing stages. User actions are passed to the IRSystem for coordination.

2. IRSystem

The IRSystem acts as the central controller and coordinator of the entire application. It follows the Facade design pattern and connects all major subsystems such as document management, query processing, retrieval, storage, and evaluation. The class manages the complete IR workflow from document import to search result generation.

Document Management Components

3. DocumentManager

The DocumentManager manages the document collection and coordinates document-level operations. It is responsible for maintaining document objects and orchestrating parsing, preprocessing, and indexing workflows.

4. Document

The Document class represents an individual document stored in the IR system. It maintains multiple versions of the document text, including the original text, tokenized text, stop-word-removed text, and stemmed text. This enables the system to display documents after different processing stages.

5. DocumentParser

The DocumentParser reads an input file and splits it into individual documents based on predefined criteria such as separators, tags, or empty lines. It converts raw input data into structured Document objects.

Preprocessing Components

6. Preprocessor

The Preprocessor represents a preprocessing abstraction applied to documents before indexing or retrieval. It provides a common contract for techniques such as tokenization, stop-word removal, and stemming.

7. Tokenizer

The Tokenizer breaks document text into smaller units called tokens or terms. Tokenization is the first preprocessing step and prepares text for stop-word removal and stemming.

8. StopWordRemover

The StopWordRemover removes common non-informative words such as “the”, “is”, “and”, and “of”. Removing stop words reduces index size and improves retrieval efficiency.

9. Stemmer

The Stemmer reduces inflected or derived words to their stem form. This improves matching between related document terms and user queries.

Indexing Components

10. Indexer

The Indexer builds the searchable index from processed documents. It extracts terms and related posting information from documents and stores them in the indexing structure used for retrieval.

11. InvertedIndex

The InvertedIndex is the primary indexing structure used by the IR system. It maps terms to postings lists containing the documents in which those terms occur and, depending on the implementation, may also store additional information such as term frequency or positions.

Query Processing Components

12. QueryProcessor

The QueryProcessor analyzes and parses raw user queries. It extracts search terms and, when supported, interprets operators such as AND, OR, and NOT before converting the input into a structured Query object.

13. Query

The Query class stores the processed representation of a user query. It may contain normalized query terms, operators, and other metadata required for search evaluation.

Retrieval Components

14. RetrievalEngine

The RetrievalEngine controls the searching process. It selects the active retrieval model, submits queries for evaluation, and returns matching documents or ranked results depending on the retrieval approach.

15. RetrievalModel

The RetrievalModel is an abstract class defining the general behavior of retrieval models. Different retrieval techniques inherit from this class and implement their own matching or ranking logic.

16. BooleanModel

The BooleanModel retrieves documents using Boolean logic. Queries are evaluated using operators such as AND, OR, and NOT, and the model returns documents that satisfy the specified logical conditions.

17. VectorSpaceModel

The VectorSpaceModel retrieves and ranks documents based on similarity calculations between query vectors and document vectors. It typically uses weighting schemes such as TF-IDF and similarity measures such as cosine similarity.

Boolean Search Implementations

18. BooleanImplementation

The BooleanImplementation interface defines a common search operation for Boolean retrieval methods. Different Boolean search techniques implement this interface using different evaluation strategies.

19. LinearSearch

The LinearSearch implementation sequentially scans every document in the collection to evaluate Boolean queries. It is simple but inefficient for large datasets.

20. InvertedListSearch

The InvertedListSearch implementation uses the inverted index to efficiently retrieve documents containing query terms. It is significantly faster than linear search.

21. SignatureSearch

The SignatureSearch implementation uses binary signatures to filter candidate documents before detailed query evaluation. This can reduce the number of full comparisons, although candidate filtering may still admit false positives that require verification.

Ranking Components

22. Ranker

The Ranker orders retrieved documents according to relevance scores or other ranking criteria. Ranking improves result usefulness by presenting the most relevant documents first when the retrieval model supports ordered output.

23. SearchResult

The SearchResult class represents a retrieved document together with associated result metadata, such as a relevance score or match information. It is used to present retrieval results to the user.

Support Components

24. StorageManager

The StorageManager handles saving and loading documents, indexes, and processing states to and from secondary storage. This enables the system to preserve data after restarting.

25. EvaluationManager

The EvaluationManager evaluates retrieval effectiveness using standard IR metrics such as precision and recall. It compares system results against a relevance ground truth for a set of evaluation queries.

- Precision: Measures the proportion of retrieved documents that are relevant.
- Recall: Measures the proportion of relevant documents that are successfully retrieved.